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Explainable AI for education: Enhancing essay scoring via rubric-aligned chain-of-thought prompting

Wenbo Xu [*,†,¶] M. S. S. Kassim [*,‖,§§] Wai Lam Hoo [*,**,§§]
Wudao Yang [‡,††] and Tianrong Xu [§,‡‡]

[*] *Faculty of Computer Science and Information Technology*
Universiti Malaya, Kuala Lumpur 50603, Malaysia

[†] *Faculty of Digital Arts*
Jiangxi Arts and Ceramics Technology Institute
Jingdezhen 333001, P. R. China

[‡] *School of Mathematics and Computer Science*
Yunnan Minzu University, Kunming 650504, P. R. China

[§] *School of Management (School of Accessibility Management)*
Nanjing Normal University of Special Education
Nanjing 210038, P. R. China
[¶] *xuwenboxue_xwb@163.com*
[‖] *shahreeza@um.edu.my*
[**] *wlhoo@um.edu.my*
[††] *wudaoyang@ymu.edu.cn*
[‡‡] *Tianrong@njts.edu.cn*

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Automated essay scoring (AES) systems increasingly leverage fine-tuned large language models (LLMs) to enhance scoring accuracy and feedback generation. However, current LLM-based AES approaches often lack interpretability and consistent alignment with human scoring rubrics, limiting their practical adoption in educational settings. This study proposes QwenScore+, a novel framework that integrates rubric-aware Chain-of-Thought (CoT) prompting with reinforcement learning from human feedback (RLHF) to improve the transparency, quality and educational alignment of automated feedback. QwenScore+ is evaluated on a proprietary IELTS writing dataset comprising over 5000 essays annotated with trait-level scores and expert-written feedback. Experimental results demonstrate that QwenScore+ significantly outperforms strong baselines such as BERT, GPT-3.5 and GPT-4 in feedback generation, achieving higher BLEU, ROUGE-L and cosine similarity scores, alongside improvements in trait-level scoring measured by quadratic weighted kappa (QWK). Furthermore, rubric-aligned CoT prompting enables the

[§§] Corresponding authors.

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generation of feedback that better mirrors human reasoning patterns, as confirmed through automatic metrics and human evaluations. These findings highlight the potential of combining explainable reasoning strategies with human-aligned reward optimization to develop more transparent, reliable and pedagogically valuable AES systems for real-world applications.

Keywords: Automated essay scoring (AES); large language models (LLMs); chain-of-thought prompting; rubric alignment; explainability.

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I. Introduction

The advent of explainable AI (XAI) in education highlights the requirement for transparency in intelligent systems. Explainability in high-stakes tasks such as essay scoring is critical in developing teacher trust, enabling actionable student feedback and aligning model output with rubric-guided learning goals.[1–4] Automated essay scoring (AES), a core task in educational natural language processing (NLP), has evolved from traditional feature-based models to deep neural architectures such as BERT,[5] and more recently to fine-tuned large language models (LLMs) including GPT-3.5 and GPT-4.[6–8] While these models demonstrate superior contextual understanding and scoring performance — typically measured by metrics such as quadratic weighted kappa (QWK)[9,10] — they often function as black-box systems, providing limited insight into the rationale behind their outputs.[10–12]

Recent advances in intelligent classroom technologies have facilitated the integration of AES within real-time digital learning platforms. For instance, Liu *et al.* proposed a graph generative diffusion model with hard negative sampling for enhanced recommendation performance,[13] while Liu *et al.* introduced an influence maximization approach using graph attention weight fusion (GAWF),[14] showcasing the potential of graph-based methods in complex educational contexts. As illustrated in Fig. 1, student essays can be automatically evaluated and scored upon submission using LLM-based systems.[7,15]

Despite their promise, current AES models face two central challenges: limited interpretability and the lack of fine-grained annotated data. Most publicly available datasets provide only coarse-level scores and omit detailed, trait-specific feedback,

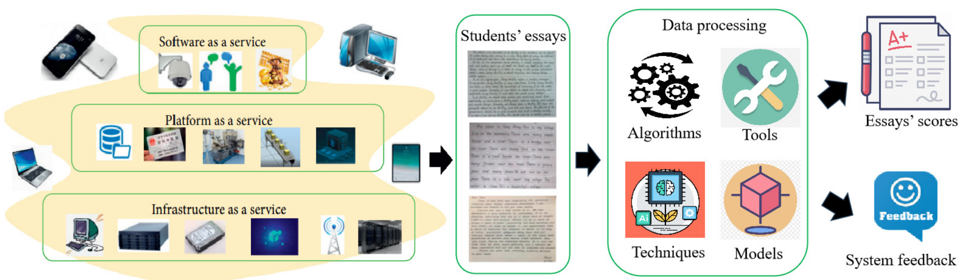

Fig. 1. Automated essay scoring workflow in digital classroom settings.

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constraining both model supervision and explainability.[9,16] Furthermore, annotation inconsistencies across existing datasets hinder generalization and reproducibility.[7]

To address these limitations, this study develops a proprietary IELTS essay dataset consisting of over 5000 student responses, each annotated by certified IELTS writing examiners in accordance with the standardized Task 2 Band Descriptors (see Appendix A).[11,17,18] Each sample includes trait-level scores and paragraph-level feedback covering the four core dimensions: task response (TR), coherence and cohesion (CC), lexical resource (LR) and grammatical range and accuracy (GRA). The International English Language Testing System (IELTS) is widely adopted in academic and professional contexts to evaluate English proficiency in nonnative speakers, making it an ideal benchmark for applied AES research.

Building upon this dataset, we introduce QwenScore+, a novel AES framework that integrates rubric-aware chain-of-thought (CoT) prompting and Reinforcement learning from human feedback (RLHF), implemented on the Qwen language model — a bilingual Chinese-English LLM developed by Alibaba.[17,19,20] The CoT mechanism imitates the step-by-step reasoning of a human rater by breaking down holistic judgments into structured, trait-specific explanations consistent with the rubrics in IELTS. This reasoning trace enhances the transparency and explainability of scoring choices. Unlike traditional, opaque black-box AES models, our approach provides criterion-referenced explanations supporting pedagogical knowledge and evaluative confidence. To our knowledge, this represents the first integration of rubric-aligned CoT prompting into Qwen for essay scoring and feedback generation. While CoT has proven effective in tasks such as arithmetic reasoning and question answering,[21,22] its application to educational assessment remains largely unexplored.

The proposed system decomposes evaluation tasks into intermediate reasoning steps that are explicitly aligned with rubric descriptors, thereby enhancing the traceability and clarity of model decisions.[23] Further, by incorporating RLHF during fine-tuning, QwenScore+ is guided toward generating feedback and scores that closely resemble human raters' preferences.[24,25]

The key contributions of this work are as follows:

(i) *Construction of a rubric-aligned IELTS dataset with expert feedback:* We introduce a new AES benchmark comprising over 5000 IELTS essays with trait-level scores and paragraph-level comments, enhancing both training quality and model interpretability.
(ii) *Rubric-aware CoT prompting in Qwen-based AES:* Our system incorporates structured prompts that embed human scoring rubrics, enabling trait-specific reasoning and rubric-consistent feedback generation.
(iii) *RLHF-enhanced interpretability and alignment:* By coupling rubric traits with reasoning traces and optimizing via RLHF, QwenScore+ produces coherent, criterion-referenced outputs that advance transparency and trustworthiness in AES.

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2. Related Work

This section reviews the relevant literature across three key dimensions: (1) the integration of XAI into AES, (2) the development of CoT prompting mechanisms and (3) the design of rubric-aligned scoring strategies. While each area has witnessed notable progress in recent years, existing research still exhibits critical gaps in applying these methods to educational assessment contexts. The following subsections analyze the advancements and limitations in these domains, and identify the unexplored intersections that this study aims to address.

2.1. *Research and challenges of XAI in AES*

In recent years, XAI has increasingly been incorporated into educational testing in order to make automated scoring more transparent and consistent. Some studies have introduced methods based on score visualization, feature attribution and attention-based interpretability in an attempt to map the decision-making of scoring models. These methods, however, have relied heavily on traditional feature engineering or simple neural architectures, and as such are limited in describing complex linguistic concepts entailed in essay assessment. Adversely, the use of fixed scoring logic in rule-based systems provides little flexibility for variation in writing styles or test prompts. *Post-hoc* XAI methods like LIME or SHAP, on the other hand, have proved poorly aligned with rubric assessment since they optimize for feature attribution over criterion-reference scoring.[10]

More recent efforts have focused on enabling rationale-based AES. For instance, Chu *et al.*[23] proposed a framework combining scoring-LLMs (S-LLMs) with generative explanations to support multi-trait scoring. Similarly, Zhang *et al.*[21] proposed an automatic CoT prompting approach that integrates intermediate reasoning steps into LLMs, offering a promising direction for enhancing interpretability and alignment in educational NLP tasks such as AES. While such studies have advanced the application of XAI in AES, they primarily focus on output-level interpretability and often lack the systematic integration of rubric criteria, feedback strategies and internal reasoning paths.

2.2. *Progress and transfer gaps in CoT reasoning mechanisms*

CoT prompting has been widely applied in domains such as mathematical reasoning, logical QA and commonsense inference. Its core strength lies in explicitly exposing the step-by-step reasoning process, thereby improving the accuracy and interpretability of model outputs.[19,20] For example, Cao *et al.*[17] proposed the Tree-of-Thought framework, which supports branching decision pathways and excels in complex reasoning tasks. Diao *et al.*[24] further optimized reasoning consistency through active prompting strategies.

Despite these advancements, the application of CoT in educational assessment — particularly in AES — remains largely unexplored. No existing study has systematically

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employed CoT as a scoring inference mechanism to guide feedback generation aligned with rubric-based evaluation. This work pioneers the integration of CoT into the essay scoring pipeline, further enhanced by RLHF, to construct human-preference-aligned reasoning paths for scoring.

2.3. *Exploration and limitations of rubric-aligned scoring mechanisms*

Rubric-based scoring, also called rubric-guided assessment, involves assessing essays according to certain explicitly defined criteria as stipulated in fixed rubrics for scoring. For Task 2 writing on the IELTS, this consists of four general dimensions, including TR, CC, LR and GRA. These dimensions constitute a consistent and interpretable framework for both human and automated assessors.

Early versions of the AES used hand-humanated features in order partly to align scoring with such rubric criteria, relying on principled logic and surface-level linguistic feature extraction.[26] Recent approaches have instead tried to incorporate rubric dimensions into neural network architectures or prompt forms in order partly to enhance alignment, scoring consistency and control.

For example, Hashemi and Zhang[12] introduced LLM-R UBRIC, a calibrated framework for aligning language model scoring with multidimensional rubrics. Henkel *et al.*[18] incorporated rubric-aware CoT prompting in the AMMORE dataset to enhance assessment granularity, particularly in handling edge cases. While these studies show promise in rubric integration, most fail to tightly couple rubric logic with interpretable inference mechanisms, thus limiting their capacity to produce cognitively aligned feedback and scores.

3. Datasets and Methods

This section outlines the datasets used in this study and the methodological framework adopted to integrate CoT reasoning into AES. We first introduce the publicly available datasets and our proprietary IELTS corpus, which provides fine-grained, expert-annotated data for both scoring and feedback generation. We then present the CoT-enhanced scoring and feedback framework, detailing how reasoning steps, scoring criteria and feedback strategies are systematically embedded into the model's prompting and generation process.

The following figure provides an overview of the entire methodological pipeline adopted in this study. Figure 2 illustrates the main components, including dataset input, rubric-aware prompting, structured CoT reasoning and final score and feedback generation stages.

3.1. *Public datasets and IELTS private dataset*

Finding suitable datasets is critical for advancing AES effectiveness, as they provide essential data for model evaluation. For an AES system to accurately evaluate an

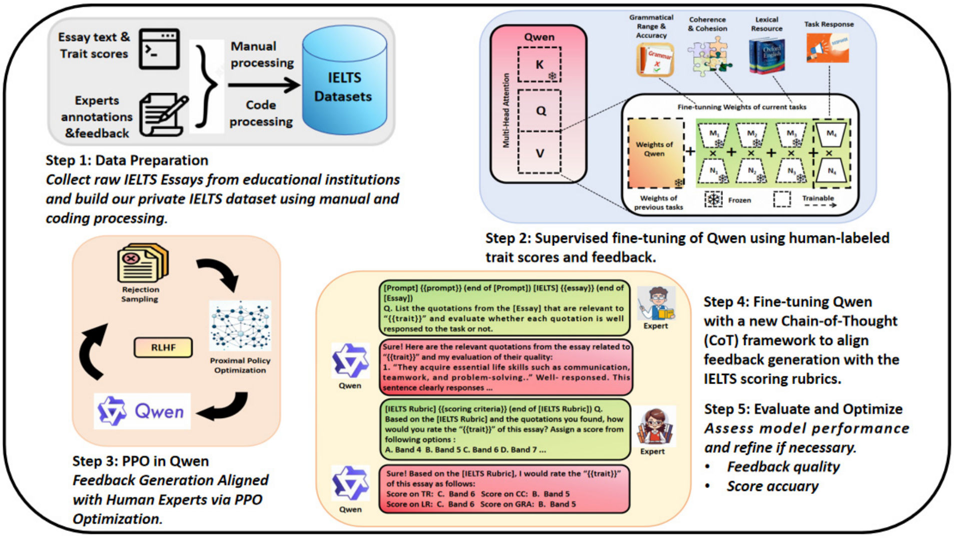

Fig. 2. Overview of the proposed methodology integrating CoT reasoning into AES for structured scoring and feedback generation.

essay, it must be trained to recognize patterns in the data and to understand the characteristics of good and bad essays. The datasets used to train and evaluate an AES system must include various essays, including essays of different lengths, different topics and written by different authors. This will ensure that the system can properly identify and evaluate the various types of essays that it will be asked to assess. To the best of our knowledge, seven English composition datasets are widely accepted and used in the AES field. Their details are demonstrated in Table 1.

Table 1. Commonly used public datasets in automated essay scoring.

Dataset	Writer	Samples	Prompts	Scoring	Rubric	Annotations
CLC-FCE[a]	Nonnative	1244	10	Holistic (1–40)	ESOL	80 error types
EFCAMDAT[b]	Nonnative	1 180 310	1447	Holistic (A1–D4)	CEFR	Grammar, POS
TOEFL11[c]	Nonnative	12 100	8	Holistic (Low/Med/High)	TOEFL	None
SciEntsBank[d]	Native	15 357	15	Holistic (Unaddressed)	FOSS	145k facet entailment
ASAP	—	12 980	8	Holistic (1–6)	OWSG	None
ASAP-SAS	Native	27 588	10	Holistic (0–2/3)	SOEI	None
ASAP++[e]	—	10 688	6	Attribute-based (4/5)	OWSG	None
ICLEv3[f]	Nonnative	9529	258	Holistic (B2/C1/C2)	CEFR	POS tagging
DREsS[g]	Nonnative	1782	22	Holistic (1–5) & Traits	EFL	None

Notes: [a]Cambridge Learner Corpus (FCE): https://www.ilexir.co.uk/datasets/index.html.
[b]EF-Cambridge Open Language Database: https://ef-lab.mmll.cam.ac.uk/EFCAMDAT.html.
[c]TOEFL11: https://catalog.ldc.upenn.edu/LDC2014T06.
[d]SciEntsBank: https://huggingface.co/datasets/nkazi/SciEntsBank.
[e]ASAP: https://www.kaggle.com/c/asap-aes, SAS: https://www.kaggle.com/competitions/asap-sas/data, ASAP++: https://cfilt.iitb.ac.in/~egdata/.
[f]ICLEv3: https://corpora.uclouvain.be/cecl/icle/trial/.
[g]DREsS: https://arxiv.org/pdf/2402.16733.pdf.

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In some studies, researchers refer to the datasets they use as corpora. For convenience, we will refer to this concept uniformly as datasets in this paper.

Although publicly available datasets like ASAP or TOEFL11 are useful for model assessment, they have drawbacks such as coarse-grained scores, limited paragraph-level feedback, inconsistent annotation techniques and little manual quality checking. These drawbacks limit the construction of explainable and rubric-congruent AES models — particularly in educational use, where fine-grained analysis at the level of traits and cognitively valid feedback are essential. Even the majority of existing datasets lack expert-verified annotations and inter-rater agreement checks, thereby lowering the validity of the datasets for high-stakes assessment situations.

To address these gaps, a proprietary IELTS writing dataset was constructed and annotated as part of this study. The dataset comprises 5088 essays collected from IELTS candidates, with each essay independently scored by two certified IELTS examiners across four official traits: TR, CC, LR and GRA. In cases where trait-level score discrepancies exceeded one full band, senior adjudicators were engaged to arbitrate and ensure scoring validity.

Beyond trait scores, the dataset includes fine-grained feedback annotations aligned with the IELTS public band descriptors. Annotators were trained through a rubric calibration process, and their inter-rater reliability was evaluated using Fleiss' Kappa ($\kappa \geq 0.80$) across all traits, demonstrating strong scoring consistency. A data cleaning pipeline combining human validation and GPT-4-based prompt-engineered feedback normalization was employed to enhance the dataset's structural integrity and reduce annotation variance.

A sample JSON format of the annotated dataset is illustrated in Appendix B, providing insight into its structural design, scoring fields, and feedback schema. This IELTS dataset serves as a foundation for the model's supervised fine-tuning (SFT), reward model training and reinforcement learning with human feedback (RLHF), contributing to both scoring accuracy and interpretability in AES. Building upon this curated resource, the following section introduces the methodological framework that integrates CoT reasoning into the automated scoring and feedback generation process.

3.2. *CoT-based integration of feedback generation and scoring mechanism*

Even with impressive progress, AES systems have always struggled to provide structured and explainable feedback against human grading criteria. CoT framework decomposes difficult evaluation tasks into systematic, step-by-step sub-tasks so that scoring and generation of feedback are transparent as well as consistent with predefined IELTS writing criteria. With the application of CoT, the model assesses essays along four main dimensions: TR, CC, LR and GRA. This decomposition makes it possible to have a fine-grained assessment of essay quality, enhancing the accuracy as well as explainability of automatic scoring models.

As compared to conventional AES models returning a holistic score with no explicit reasoning procedure, this research presents a new CoT-based method that breaks down essay assessment into systematic steps. Such breakdown improves transparency as it makes it easier for automatic models to provide explanations of scores and provide more interpretable feedback.

3.2.1. *Trait-level scoring with CoT and rubric alignment*

Traditional AES systems typically return a holistic score without analyzing specific writing dimensions or offering reasoning for the assigned score. This lack of transparency undermines the pedagogical value of automated feedback and diminishes user trust. In contrast, the CoT framework enables structured, step-wise reasoning, aligning the scoring process explicitly with IELTS rubric dimensions.

Following recent progress in CoT prompting,[19] our system employs a hierarchical evaluation strategy where the scores for each of the four traits — TR, CC, LR and GRA — are assessed independently and aggregated into a final band score:

$$\mathrm{Score}_{\mathrm{final}} = f(\mathrm{Score}_{\mathrm{TR}}, \mathrm{Score}_{\mathrm{CC}}, \mathrm{Score}_{\mathrm{LR}}, \mathrm{Score}_{\mathrm{GRA}}), \tag{1}$$

where $f(\cdot)$ is a weighted aggregation function that integrates the trait-level scores.

To enhance rubric alignment and interpretability, attention-based mechanisms are incorporated to identify salient features aligned with dimension-specific descriptors. For example, the model locates evidence relevant to coherence in linking words or lexical range in vocabulary diversity. This ensures that each score is grounded in rubric criteria and supported by textual justification, improving both fairness and explainability in AES.

3.2.2. *Feedback generation via hierarchical reasoning*

The same CoT-based reasoning structure is extended to generate actionable, paragraph-level feedback. Instead of relying on post-hoc explanations, the model produces feedback as a by-product of its trait-level evaluation, enhancing both interpretability and pedagogical relevance.

The reasoning pipeline consists of the following stages:

- Feature Extraction: Identifying linguistic and semantic patterns relevant to each scoring dimension.
- Rubric Mapping: Aligning extracted features with the IELTS public band descriptors to ensure rubric compliance.
- Hierarchical Justification: Structuring reasoning paths that explain how essay characteristics justify trait-level scores.
- Actionable Feedback Generation: Translating justifications into specific, interpretable suggestions that guide student improvement.

By integrating structured reasoning into the feedback pipeline, the system avoids generic comments and instead provides personalized guidance tied to each dimension.

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Attention-based alignment further enhances relevance by focusing on linguistically and structurally salient components of the essay. This not only improves model transparency but also ensures that learners receive feedback grounded in the same evaluative logic used for scoring.

3.2.3. *Implementation of CoT in the scoring model*

A multi-step logical reasoning framework is utilized to deploy CoT within an AES scheme that incorporates a transformer-based CoT decoder alongside QWEN. It adopts a pipeline architecture with a number of crucial elements:

- Feature Extraction Layer: Fine-tuned to extract linguistic features from essays, highlighting key elements within the four IELTS scoring dimensions.
- Rubric Alignment Layer: This layer applies attention-based matching between extracted features and IELTS rubrics, consistently applying evaluation criteria.
- Logical Reasoning Layer: The CoT reasoning framework is used to infer step-by-step justifications for each assigned score, making the scoring process transparent and interpretable.
- Feedback Generation Module: The system generates personalized, structured feedback using fine-tuned reinforcement learning strategies.

Mathematically, the rubric alignment score S_d for dimension d is computed as

$$S_d = \sigma(W_d \cdot \mathrm{Attention}(Q_d, K_d, V_d)), \tag{2}$$

where W_d represents task-specific learnable parameters, and Q_d, K_d, V_d are query, key and value representations used in the attention mechanism for rubric alignment.

To further refine feedback quality, this study incorporates proximal policy optimization (PPO), a reinforcement learning strategy, to enhance feedback alignment with human expert evaluations. Given a set of expert-annotated reference feedback samples F^*, the reward function maximizes alignment between model-generated feedback F and human references:

$$R(F, F^*) = \mathrm{cosine}(\mathrm{Embed}(F), \mathrm{Embed}(F^*)). \tag{3}$$

By iteratively fine-tuning the model with PPO updates, the system refines its ability to generate concise, informative and personalized feedback aligned with expert assessments.

Together, these components establish a foundation for reliable scoring and personalized feedback generation in AES systems.

3.2.4. *Algorithmic implementation of the CoT framework*

The CoT framework logically structures the essay feedback and scoring process, ensuring systematic evaluation aligned with IELTS scoring standards. Algorithm 1 outlines the logical steps for CoT-based structured scoring and feedback generation.

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Algorithm 1. Enhanced CoT-based essay feedback and scoring using QWen LLM

```
Input: IELTS essay text
Output: Feedback, Scores and Confidence Levels for TR, CC, LR and GRA
Initialize QWen LLM model: QWen_LLM
for each essay do
    Step 1: Preprocess essay text (tokenization, normalization)
    Step 2: Generate detailed feedback for each dimension (TR, CC, LR, GRA)
    for each dimension in {TR, CC, LR, GRA} do
        Extract feedback from generated text for the specific dimension
    end for
    Step 3: Compare feedback with Band Descriptors
    for each dimension in {TR, CC, LR, GRA} do
        Compute weighted similarity with band descriptors
        Assign score based on highest similarity
        Compute confidence score
        if confidence score ¡ threshold then
            Re-evaluate with an alternative prompt
        end if
    end for
    Step 4: Output final scores and store reasoning steps
    Store final scores, confidence levels and reasoning logs
end for
```

This algorithmic methodology provides a systematic and open evaluation procedure. By breaking down scoring and feed-forward generation into explicit steps, the CoT approach improves the reliability and explainability of AES models. Empirical results show that the CoT-augmented scoring system surpasses that of conventional AES models with regards to scoring reliability and feed-forward quality by achieving superior QWK and BLEU scores as well as superior human assessment ratings. Merging logical reasoning, rubric alignment, as well as reinforcement learning methods assures that feed-forward from automatic scoring resembles that of human expert assessments, finally enhancing the educational value of automatic essay scoring methods.

3.3. *Evaluation metrics*

To evaluate the relevance and quality of the provided feedback, this work utilized a blend of lexical overlap as well as semantic similarity measures. It chose these measures due to their effectiveness as tested for natural language generation tasks as well as their applicability to measuring the quality of educational feedback.

Specifically, three metrics were adopted:

- BLEU[27]: Measures n-gram precision between generated and reference feedback, capturing lexical fluency and local accuracy.

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- ROUGE-L[28]: Evaluates the longest common subsequence between texts, reflecting content recall and structural consistency.
- Cosine Similarity: Computes the semantic similarity between feedback embeddings.

$$\text{Cosine similarity} = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\|\|\mathbf{B}\|}, \tag{4}$$

where $\mathbf{A}$ and $\mathbf{B}$ denote the embedding vectors of the model-generated feedback and the human expert feedback, respectively. The numerator $\mathbf{A} \cdot \mathbf{B}$ represents the dot product between the two vectors, while the denominator $\|\mathbf{A}\|\|\mathbf{B}\|$ is the product of their Euclidean norms.

The incorporation of these measures facilitates the holistic assessment that captures correspondence as well as semantic alignment at a deeper level between model-provided feedback and human references. Such strategies of evaluation align perfectly with the goals of rubric-based and pedagogically rich feedback generation as pursued through this research.

Aside from automatic scoring, a human evaluation procedure was performed. IELTS specialists rated the feedback according to a five-point Likert scale of relevance, specificity and clarity as common practices for subjective quality assessment.[29] Additionally, pairwise preference ratings were gathered for identification of which versions of the feedback were rated as more educationally beneficial, using comparative evaluation techniques as found in standard human alignment tasks for feedback.[30] This qualitative assessment serves as a supplement to automatic measures, offering a holistic quality estimation of the feedback.

4. Experimental Settings and Results

This section presents the experimental configuration and key findings related to the evaluation of the proposed AES framework. It includes implementation details, model comparison results and both automatic and human evaluation outcomes. The experiments are designed to assess trait-level scoring accuracy, feedback quality and the impact of key components such as CoT reasoning, PPO and model scaling. Each subsection corresponds to a distinct experimental focus, contributing to a comprehensive understanding of the system's effectiveness.

4.1. *Experimental environment and parameter settings*

The QWEN-based models used in this study were implemented using the PyTorch 2.0.1 framework, supported by the HuggingFace Transformers and PEFT libraries to enable parameter-efficient fine-tuning. All experiments were conducted on NVIDIA A800 GPUs with 80GB of VRAM per unit.

For SFT and inference tasks, a batch size of 16 was adopted to accommodate GPU memory constraints. The AdamW optimizer was used with an initial learning rate of 2×10^{-5}, which decayed by a factor of 0.1 upon stagnation in validation loss.

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Training was capped at 50 epochs, with an early stopping strategy employed to preserve the model checkpoint with the lowest validation loss.

The QWEN model backbone comprises 32 transformer layers, a hidden size of $H = 4096$, 32 attention heads and a feed-forward dimension of 11 008. Input token lengths were set to 512 for scoring tasks and 1024 for feedback generation. Output sequences were truncated to 150 tokens to ensure concise and interpretable feedback.

For reinforcement learning fine-tuning, the PPO algorithm was used. Hyperparameters were set as follows: clipping parameter $\epsilon = 0.2$, generalized advantage estimation $\lambda = 0.95$ and discount factor $\gamma = 0.99$. The policy and value networks used learning rates of 1×10^{-5} and 5×10^{-5}, respectively. The reward signal was computed via cosine similarity between generated feedback and reference feedback vectors, measured against rubric-aligned quality dimensions.

The above hyperparameters were selected through a combination of prior literature guidance and empirical validation. Specifically, PPO settings were adapted from RLHF practices in LLM alignment tasks,[31] while learning rates and optimization strategies were initially adopted from recent studies in LLM-based AES systems.[6,12] Grid search was performed on a validation subset of 500 IELTS essays, using QWK for scoring and BLEU for feedback generation as the selection criteria. This process ensured robust convergence and maximized alignment with human preferences.

All experiments were repeated across 10 independent runs, and average scores were reported to mitigate performance variance. Baseline models such as LLaMA-3 and GPT-4 followed their respective open-source hyperparameter configurations, with prompt length and token limits standardized for fair comparison. Full prompting strategies and instructions for QWEN, LLaMA and GPT-4 models are provided in Appendix C.

4.2. *Trait-level scoring performance with the CoT framework*

For the trait-level scoring task incorporating the CoT framework, the model outputs dimension-specific predictions across the four IELTS traits: TR, CC, LR and GRA. QWK was used as the primary metric to assess alignment with human annotations, while mean absolute error (MAE) and root mean square error (RMSE) quantified the trait-level prediction deviation. These complementary metrics enable fine-grained evaluation of scoring consistency and accuracy.

As visualized in Fig. 3, the original CoT-based model demonstrated superior performance, particularly in CC (QWK $= 0.6814$, MAE $= 0.2600$, RMSE $= 0.5099$) and GRA (QWK $= 0.5640$). These results underscore the advantage of incorporating explicit reasoning steps, which appear especially beneficial for traits requiring discourse-level understanding and syntactic complexity.

In contrast, the LLaMA-3 baseline without CoT reasoning yielded the weakest performance across all traits. Notably, the QWK scores for CC and LR dropped to 0.0378 and 0.0871, respectively, and all trait-level MAE and RMSE values

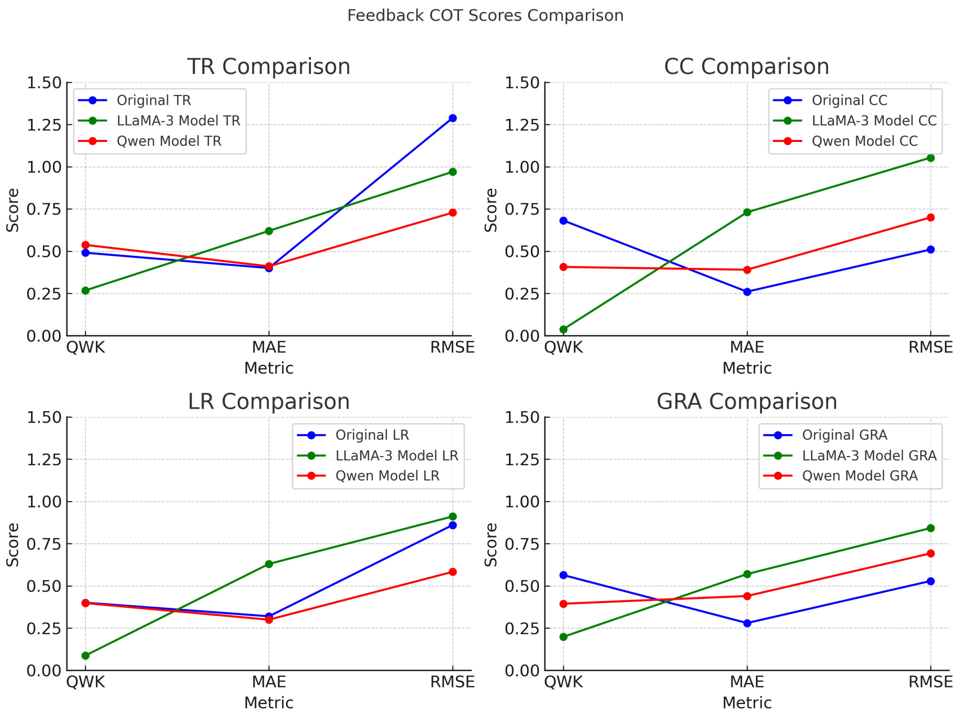

Fig. 3. Comparison of model performance across scoring dimensions using CoT framework.

increased significantly. These results demonstrate the limitations of end-to-end holistic prediction without structured trait-specific inference.

The Qwen variant incorporating both CoT and rubric-alignment attention exhibited trait-specific improvements. It achieved the highest QWK in TR (0.5372) and the lowest MAE and RMSE in LR (0.3000 and 0.5831, respectively), while CC and GRA showed only moderate gains. This suggests that rubric-guided attention mechanisms may be more effective for certain traits, such as lexical quality, than for others.

To examine whether larger models yield better trait-level predictions, Qwen-7B, 14B and 72B were compared under identical CoT and rubric-alignment settings. As shown in Table 2, Qwen-72B achieved slightly higher scores, but improvements were marginal. This suggests that in data-constrained scenarios, model scale alone does not guarantee significant gains, highlighting the importance of annotated data quality and task-specific alignment.

Table 2. Comparison of different Qwen model versions in trait-level scoring performance.

Model variant	Avg. QWK	Avg. MAE	Avg. RMSE
Qwen-7B (CoT only)	0.5253	0.3750	0.7851
Qwen-14B (CoT + Rubric Attention)	0.5372	0.3602	0.7726
Qwen-72B (CoT + Rubric Attention)	0.5410	0.3561	0.7685

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Table 3. Impact of PPO and CoT on feedback quality.

Model variant	BLEU	ROUGE-L	Cosine similarity
Baseline (no PPO, no CoT)	0.2371	0.4012	0.6125
+ PPO only	0.2698	0.4265	0.6480
+ CoT only	0.2535	0.4321	0.6601
+ PPO + CoT	**0.2873**	**0.4588**	**0.6856**

These findings collectively validate the effectiveness of CoT reasoning in improving trait-level scoring performance. In particular, CoT-enhanced models showed strong alignment with human ratings in coherence and grammar-related traits. While rubric alignment and model scaling provide incremental benefits, the most substantial improvements stem from incorporating explicit, structured reasoning aligned with scoring rubrics.

4.3. *Feedback generation performance: Impact of PPO and CoT*

To assess the contributions of PPO and CoT reasoning in feedback generation, ablation experiments were conducted. The baseline system lacked both PPO and CoT mechanisms, while other configurations introduced either or both techniques. As shown in Table 3, the combination of PPO and CoT achieved the highest scores across all automatic metrics: BLEU, ROUGE-L and embedding-based cosine similarity. PPO alone contributed to alignment with human preferences, while CoT enhanced the linguistic richness and trait-specific clarity of the feedback — particularly in coherence and grammar-related aspects.

Beyond automatic evaluation, a human assessment was conducted to evaluate the educational usefulness and clarity of the generated feedback. IELTS instructors rated feedback using a five-point Likert scale and provided pairwise preference judgments between model outputs. As visualized in Fig. 4, both Likert scores and preference rates increased consistently from the baseline to the PPO + CoT configuration, highlighting the pedagogical value of combining structured reasoning with reinforcement optimization.

These results collectively suggest that integrating reinforcement learning and CoT reasoning significantly enhances both the objective quality and subjective helpfulness of generated feedback, contributing to more interpretable and learner-aligned AES outputs.

4.4. *Effect of Qwen model size on feedback generation*

To further investigate the impact of model capacity on feedback generation quality, additional experiments were conducted using three versions of the Qwen model family: Qwen-7B, Qwen-14B and Qwen-72B. All models were trained using the same CoT prompting and PPO fine-tuning framework, under identical data and parameter settings, to isolate the effect of model size.

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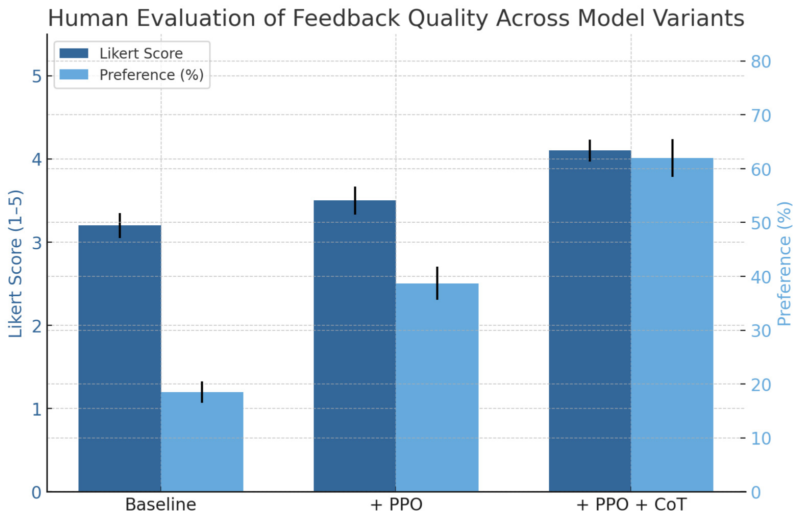

Fig. 4. Human evaluation of feedback quality across model variants. The left axis shows the average Likert score (1–5), and the right axis shows the preference percentage. Error bars represent 95% confidence intervals.

As shown in Table 4, increasing model size resulted in only marginal improvements across BLEU, ROUGE-L and cosine similarity scores. The largest model, Qwen-72B, achieved only a 2.5% improvement in BLEU and a 0.6% improvement in cosine similarity compared to Qwen-7B. This performance plateau suggests that under limited data conditions, the benefits of scaling up model size are constrained.

This observation aligns with prior studies on model scaling laws, which emphasize the interdependence between model size, data availability and compute resources.[32,33] In the context of automated feedback generation for AES, where high-quality annotated feedback is scarce and costly to obtain, increasing model capacity alone does not yield substantial quality gains.

These findings reinforce the importance of investing in high-quality, diverse and representative training datasets for effective model scaling in educational NLP applications. Particularly in specialized tasks such as IELTS feedback generation, data-centric approaches may offer more practical and interpretable improvements than scaling parameters alone.

Table 4. Feedback generation quality across Qwen model versions.

Model variant	BLEU	ROUGE-L	Cosine similarity
Qwen-7B (CoT + PPO)	0.2873	0.4588	0.6856
Qwen-14B (CoT + PPO)	0.2931	0.4622	0.6880
Qwen-72B (CoT + PPO)	0.2945	0.4630	0.6891

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5. Discussion

This section synthesizes key findings from the experimental evaluation of QWEN-Score+, focusing on its interpretability, feedback quality and architectural contributions. The discourse presents how the application of CoT reasoning, parameter-efficient fine-tuning and reinforcement learning improves the performance as well as the understandability of automatic essay scoring techniques. This section does not confine experimental conditions but relates theoretical insights to educational application implications.

5.1. *CoT-based reasoning and interpretability*

One of the fundamental design features of QWENScore+ is its application of CoT reasoning. In contrast with direct regression models, CoT architecture informs the model to proceed with structured steps: essay content analysis, generating rubric-aligned feedback and logical inference of trait-specific scores. This modular design improves scoring accuracy as well as interpretability.[19,20]

By using CoT, the model can produce middle-out rationales tracing the decision-making progress across dimensions. Such rationales are open explanations, mapping the model's output to people's assessment logic for enhancing explainability, usability, as well as faith with pedagogical applications.

Interestingly, CoT models performed the best in CC and TR — dimensions most dependent on discourse structure as well as content relevance — indicating that interpretability as well as accuracy can be attained together. Additionally, rubric-sensitive mechanisms within the Qwen variant even anchored trait scoring more firmly to semantically pertinent cues.

Relative to baseline models like GPT-4 or vanilla LLaMA-3 prompting, QwenScore+ reliably outperforms in scoring consistency as well as feedback alignment. Its better performance stems from three reasons: (1) *rubric-aware CoT prompting* imposing structured, dimension-specific reasoning under IELTS guidelines; (2) *consistent trait-level scoring* over a range of disparate prompts, facilitated by modular reasoning and pre-training for alignment; and (3) *human feedback reinforced learning* (*RLHF*), refining the model's bias toward human-rated, pedagogy-driven outputs. These combined forces help QwenScore+ produce interpretable, rubric-aligned and contextually appropriate feedback with increased scoring consistency.

5.2. *Feedback structuring via CoT and rubric alignment*

Beyond quantitative gains, qualitative gains were also seen within the structure of the feedback with the inclusion of the CoT framework.[34] By making the model reason step-by-step, from essay strength and weakness identification to mapping against trait-specific criteria, CoT provided a more logical, pedagogically consistent structure to the generated feedback.

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This structured rationale method greatly enhanced the interpretability of the feedback, especially for educational purposes where precision and rubric consistency are paramount. Not only was the resulting feedback more organized but also more consistent with the IELTS assessment framework, as indicated by enhanced ROUGE-L and Cosine Similarity scores.

Additionally, CoT and PPO had complementary effects: CoT organized the internal reasoning flow structure, while PPO fine-tuned the output for alignment with human preferences. Combined, these elements improved the objective quality as well as subjective acceptability of the generated feedback, as attested by human judgment.

5.3. *Interpretability through structured reasoning*

The QWENScore+ architecture uses two complementary training approaches of SFT and RLHF supplemented with a CoT reasoning mechanism. It is designed to improve the explainability and pedagogical calibration of the LLMs when they are applied for automatic scoring of essays as well as generating feedback.

SFT operates as a localized strategy of optimization targeting the four IELTS core scoring attributes: TR, CC, LR and GRA. It learns from exemplar data annotated by experts and gains task-specific knowledge. It provides feedback that is compliant with the professional rater's expectations.

The addition of CoT further improves explainability by breaking up the scoring into intermediary steps of reasoning. Instead of generating a final overall score directly, the model is directed to initially produce dimension-level feedback and subsequently project these explanations onto rubric-mapped scores. This explicit inference path benefits traceability as well as alignment with human assessors' thought processes.

Furthermore, the application of RLHF, specifically through PPO, fine-tunes these steps of reasoning through reward-based feedback.[35,36] Human preferences are utilized to reinforce desirable output patterns, allowing the model to generate feedback that is not only accurate but also contextually appropriate, clear and useful to learners.

The combined effect of CoT, RLHF and SFT demonstrates that performance and interpretability are by no means incompatible. Instead, step-wise reasoning mechanisms are able to make scores more reliable while at the same time making it even more transparent, which is a crucial factor for building faith in computer-based learning systems.

Together, the results at the level of traits, generating feedback and generalization show that the architecture of QWENScore+ gains from a holistic optimization approach. Instead of independent improvements, CoT reasoning, fine-tuning with LoRA and RLHF reinforcement learning all work together synergistically at the level of modules to promote consistency, interpretability, as well as robustness.

5.4. *Theoretical and practical implications*

The emergence of QWENScore+ holds substantial theoretical as well as practical implications for the AES community. Architecturally, the model is a hybrid model

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that combines discriminative and generative modeling. Integrating encoder-type modules (as BERT for rubric alignment) with generative LLMs (such as Qwen) enables a single scoring-feedback pipeline capable of executing both numeric evaluation as well as text-based explanation tasks.

From a training perspective, the two-stage optimization approach — SFT followed by RLHF — provides a blueprint for aligning LLM behavior with human instructional goals.[37,38] SFT ensures rubric fidelity, while RLHF encourages adaptability and refinement through human-in-the-loop supervision. This approach aligns with new directions of human-aligned language model research and creates new opportunities for educational AI application.

Practically, the efficacy of the QWENScore+ model demonstrates the feasibility of rubric-sensitive, scalable scoring for IELTS-level high-stakes testing. It provides real-time trait-level ratings, reduces the weight of grading on teachers, as well as gives students individualized feedback based on formal criteria. Its ability to mirror the weighing of learner as well as expert expectations, as shown from scores from human evaluation, makes it more effective for classroom learning as well as online learning.

Nevertheless, challenges exist. They include input length constraints of scoring rubrics, consistency of training data, as well as difficulty in fully representing advanced rubric logic using general-purpose LLMs. Resolving these will involve creating more diverse training data sets, adaptive rubric encoding methods, as well as potentially hybrid models integrating symbolic and neural reasoning.

In summary, the QWENScore+ methodology advances the theoretical foundations of interpretable scoring and feedback modeling while providing a sound basis for practical application within educational contexts. Subsequent revisions must consider fairness, reducing bias and equal access issues as well, particularly when applied to diverse student populations within real-world deployments.

6. Limitations and Future Work

While the proposed rubric-aligned CoT prompting framework demonstrates strong performance and interpretability, several limitations remain that warrant further research.

First, the generalization capability of the model across diverse prompts and rhetorical structures remains a challenge. Although CoT prompting has demonstrated improved rubric alignment, it occasionally exhibits hallucinations when processing underrepresented or atypical prompts.[19,39] These include speculative, culturally specific, or genre-divergent writing tasks that deviate from the training distribution. Moreover, essays written by learners with nonnative language constructs — such as unconventional syntax, code-switching, or regional expressions — may not be accurately evaluated due to their limited representation in the training data.

This limitation is consistent with findings by Zhao *et al.*, who emphasized that integrating both local and global information can enhance model robustness in

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complex networks.[40] Inspired by this, future work could explore multi-scale representation strategies in AES, incorporating discourse-aware training mechanisms, prompt diversification and dialect-sensitive fine-tuning. Such approaches could improve the model's robustness while promoting fairness and inclusivity in educational assessments.

Second, despite using trained annotators, minor inconsistencies in human scoring introduced noise into the training data. Refining annotation protocols and incorporating automated adjudication strategies, such as GPT-assisted scoring review, may improve label reliability and model stability.

Third, the computational cost of RLHF remains high, posing challenges for deployment in resource-limited settings.[31,41,42] Research into efficient alternatives like QLoRA, AdapterFusion, or lightweight RL algorithms could make training more scalable and accessible.

Additionally, input length limitations in current architectures hinder the integration of full-length essays and rubrics within a single inference step. Hierarchical encoding or long-context transformer models may offer practical solutions to this constraint.

Finally, the potential for embedded biases in pretraining data necessitates ongoing fairness evaluation. Future developments should incorporate bias auditing pipelines, counterfactual data augmentation and interpretability mechanisms to ensure equitable outcomes across learner populations.[43,44]

Beyond IELTS, the adaptability of QwenScore+ to other standardized writing assessments — such as TOEFL, GRE, or national academic writing tests — presents a promising direction for future exploration. While the core mechanism of rubric-aligned CoT prompting is task-agnostic, effective transfer to new domains may require rubric redefinition, domain-specific calibration and potentially additional training using aligned datasets. Techniques such as domain adaptation, prompt tuning, or multi-task learning could help facilitate generalization across different writing contexts, thereby enhancing the versatility and applicability of the framework in broader educational settings.

7. Conclusions

This study presents a comparative evaluation of rubric-aligned CoT prompting across multiple open-source LLMs for AES. Using a high-quality, expert-annotated IELTS dataset, the study demonstrates that CoT prompting significantly improves scoring interpretability and feedback quality. Among the evaluated models, Qwen showed the best alignment with human scoring criteria and pedagogical feedback standards.

The key contributions of this research are threefold: (1) proposing a rubric-aware prompting strategy that enhances reasoning transparency in AES; (2) systematically comparing multiple LLMs to assess their suitability for CoT integration; and (3) validating the approach through both automatic metrics and human evaluation.

Overall, this framework offers a promising pathway toward developing more explainable, trustworthy and pedagogically meaningful AES systems. Future research should focus on extending the method to multilingual and multimodal contexts, enabling adaptive feedback delivery and improving fairness and scalability in real-world educational environments.

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The IELTS is a globally recognized standardized test used to evaluate the English proficiency of nonnative speakers in academic, professional and immigration contexts.

Data Availability

The source code, datasets and supplementary materials are publicly available at: https://github.com/Owenxu0409/LLMs-AES-IELTS. The base LLM Qwen used in this study is open-source and accessible via the official repository: https://github.com/QwenLM/Qwen.

Conflict of Interest

The authors declare that there are no competing interests associated with this work.

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Appendix A. IELTS Writing Task 2 Rubrics — Band 5 Example

This appendix presents the detailed descriptors for Band 5 performance on the IELTS Writing Task 2 rubric, organized by scoring dimension.

Task Response (TR)

- Addresses the task only partially; the format may be inappropriate in places.
- Expresses a position but the development is not always clear and there may be no conclusions drawn.
- Presents some main ideas but these are limited and not sufficiently developed; there may be irrelevant detail.

Coherence and Cohesion (CC)

- Presents information with some organisation but there may be a lack of overall progression.
- Makes inadequate, inaccurate or over-use of cohesive devices.
- May be repetitive because of lack of referencing and substitution.
- May not always use paragraphs logically.

Lexical Resource (LR)

- Uses a limited range of vocabulary, but this is minimally adequate for the task.
- May make noticeable errors in spelling and/or word formation that may cause some difficulty for the reader.

Grammatical Range and Accuracy (GRA)

- Uses only a limited range of structures.
- Attempts complex sentences but these tend to be less accurate than simple sentences.
- May make frequent grammatical errors and punctuation may be faulty; errors can cause some difficulty for the reader.

The complete band descriptors are available at the official IELTS website: https://ielts.org/organisations/ielts-for-organisations/ielts-scoring-in-detail.

Appendix B. IELTS Dataset JSON Visualization

Table B.1 presents an example of a data entry from the annotated IELTS dataset used in this study.

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Table B.1 IELTS dataset example.

Essay ID	0001
Title	Some people claim that many things that children are taught at school are a waste of time. Other people argue that everything they study at school is useful at some time. Discuss both views and give your own opinion.
Essay	Consumerism seems impossible to escape for people living in our day and age. Now, more than ever, advertisers attempt to bombard all of our waking moments with persuasive sales pitches... In general, the more and more pervasive advertisements are depriving consumers of their free will and abducting their power of choice...
Scores	**TR**: 4, **CC**: 4, **LR**: 5, **GRA**: 5, **Overall Score**: 4.5
Annotations	**TR Positives**: Addressing the Prompt: The essay discusses both views on school education... **TR Negatives**: Depth of Analysis: The essay does not fully develop the arguments for both sides... **CC Positives**: Logical Flow: The essay attempts to maintain a logical flow with some connecting phrases... **CC Negatives**: Transitions between sentences are often awkward or missing... **LR Positives**: Range of Vocabulary: The essay demonstrates an attempt to use a range of vocabulary... **LR Negatives**: Word Choice: Some word choices are inaccurate and awkward... **GRA Positives**: Sentence Structure Variety: The essay attempts a variety of sentence structures... **GRA Negatives**: Frequent Grammatical Errors: The essay contains numerous grammatical errors...
Suggestions	**Introduction**: The introduction should be clearer and more concise... **Body Paragraph 1**: Improve the logical structure and clarity of sentences... **Body Paragraph 2**: Use clear examples and logical transitions to support your points... **Body Paragraph 3**: Discuss the counterargument with clear examples... **Conclusion**: Strengthen the conclusion by summarizing key points more effectively...

Appendix C. Prompting Instructions for LLMs

GPT-4 Zero-Shot Scoring Instructions for IELTS Writing

This Excel file contains the following columns:

- **essay_id**: A unique identifier for each essay.
- **essay_prompt**: The prompt to which the essay responds.
- **essay**: The actual content of the essay.

The task is to randomly select 1000 records from this file for scoring. Please adhere to the following guidelines:

- TR, CC, LR, and GRA scores must be integers ranging from 0 to 9.
- The Overall score should be the average of these four scores, rounded to the nearest 0.5 (for example, 6.25 should be rounded to 6.5).
- The scoring should consider both the **essay_prompt** and the **essay** content.
- Please save the **essay_id** and the original scores for the randomly selected essays, as they will be needed for comparative analysis later.

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GPT-4 Few-Shot Scoring Instructions for IELTS Writing

In a few-shot learning setting, please follow these additional guidelines along with the zero-shot scoring instructions:

- Provide exemplar essays with corresponding band scores for TR, CC, LR, and GRA. These exemplars should include:
  - **Low-Band (0–4)**: Essays that have significant issues in addressing the task, coherence, vocabulary and grammar, with frequent errors.
  - **Mid-Band (5–6)**: Essays that meet basic task requirements but have some issues with idea development, coherence and occasional errors in vocabulary and grammar.
  - **High-Band (7–9)**: Essays that fully address the task, are well-organized, use a varied vocabulary and have minimal grammatical errors.
- Ensure exemplars cover a range of prompts and include concise explanations for the assigned scores.

Task: Grade the following essay using the *IELTS Writing Task 2 Band Descriptors* and the full band range (0–9).

- **Provide an explanation for your score**: [Insert Score and Explanation]
- **Essay to be scored**: [Insert Essay Text]

LLaMA-3 Zero-Shot Scoring Instructions for IELTS Writing

This instruction set is intended for a fine-tuned LLaMA-3 model to perform automated scoring on IELTS Writing Task 2 essays in a zero-shot setting. The input data is structured with the following fields:

- **essay_id**: A unique identifier assigned to each essay.
- **essay_prompt**: The writing task prompt provided to the candidate.
- **essay**: The complete written response from the candidate.

Please follow the guidelines below when assigning scores:

- Assign integer band scores from 0 to 9 for the four IELTS scoring criteria: task response (TR), coherence and cohesion (CC), lexical resource (LR) and grammatical range and accuracy (GRA).
- Compute the **Overall score** as the arithmetic mean of the four trait scores and round to the nearest 0.5 increment (e.g. 6.25 → 6.5).
- All scores must be inferred based on both the prompt and the essay content.
- Store the predicted scores along with the corresponding **essay_id** for further evaluation and comparison.

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Qwen Fine-Tune Scoring Instructions for IELTS Writing

For fine-tuning the latest version of the Qwen model on IELTS Writing Task 2 scoring, the following instructions should be strictly followed:

- **Task Separation for Scoring and Feedback:** Fine-tune the model to treat scoring and feedback generation as distinct subtasks. This modular training structure enables Qwen to optimize for accurate evaluation and meaningful feedback independently.
- **Trait-Level Scoring and Feedback:** Configure the model to process each of the four IELTS scoring criteria — TR, CC, LR and GRA — independently. Each criterion should receive both a score and detailed feedback aligned with the IELTS rubric.
- **Scoring Format (Discrete Band Selection):** When training for scoring prediction, adopt a classification-style format with predefined options for each trait. For example:
  - **TR:** Select the most appropriate band score for the essay's task response:
    - A. Band 5
    - B. Band 6
    - C. Band 7
    - D. Band 8
  - Apply a similar discrete choice format for CC, LR and GRA.
- **Feedback Prompting Strategy:** After predicting the score for each dimension, the model should generate feedback explicitly aligned with that score level. For instance, if Band 6 is predicted for TR, the feedback must explain what aspects justify Band 6 and what improvements are necessary to reach Band 7 or above.

Task Objective: Once the model is fully fine-tuned, perform the following on each input essay:

- **Scoring:** Assign a band score (0–9) for TR, CC, LR, and GRA based on the discrete band options.
- **Feedback Generation:** Produce tailored feedback for each scoring dimension, grounded in the rubric descriptors and the essay's performance.
- **Essay to be scored:** [Insert Essay Text]
- **Provide scores, explanations, and feedback:**
  - **TR:** [Insert Score], [Insert Explanation], [Insert Feedback]
  - **CC:** [Insert Score], [Insert Explanation], [Insert Feedback]
  - **LR:** [Insert Score], [Insert Explanation], [Insert Feedback]
  - **GRA:** [Insert Score], [Insert Explanation], [Insert Feedback]

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ORCID

Wenbo Xu https://orcid.org/0009-0008-5972-0551
M. S. S. Kassim https://orcid.org/0000-0003-1271-7888
Wai Lam Hoo https://orcid.org/0009-0006-2540-8020
Wudao Yang https://orcid.org/0000-0001-8411-1450
Tianrong Xu https://orcid.org/0009-0006-8818-8870

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